

Rasel Khan Nishan

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Education

Master of Philosophy (MPhil), Microbiology

Ongoing

Jagannath University, Dhaka

- **Thesis:** Development of a Strip-Based DNA Hybridization Assay for Detection of Selected Gram-Positive Nosocomial Pathogens and Their Resistance and Virulence Markers in Bangladesh

Master of Science (MSc), Microbiology

April 2025

Jagannath University, Dhaka

- **CGPA:** 3.85/4.00
- **Thesis:** A ceRNA-based multi-omics analysis identifies a prognostic signature in Lung Adenocarcinoma

Bachelor of Science (BSc), Microbiology

June 2023

Jagannath University, Dhaka

- **CGPA:** 3.45/4.00
- **4th year project:** Molecular identification of Escherichia Coli from environmental sample

Research Experience (3+ Years)

Research Assistant

January 2026 - Present

Advanced Molecular Biology Laboratory, **Jagannath University**, Dhaka

Advisor: Associate Professor Dr. Syeda Tasneen Towhid

Project: Development of a Strip-Based DNA Hybridization Assay for Detection of Selected Gram-Positive Nosocomial Pathogens and Their Resistance and Virulence Markers in Bangladesh

Overview:

- Developing a low-cost strip-based DNA hybridization assay to detect MRSA, vancomycin-resistant enterococci, resistant *Streptococcus pneumoniae*, and toxigenic *Clostridioides difficile*, including resistance and virulence markers.
- Applying culture, antimicrobial susceptibility testing, PCR, bioinformatic probe design, membrane hybridization, and analytical validation using reference strains and local clinical isolates.

Graduate Research Assistant

July 2024 - December 2025

Department of Microbiology, **Jagannath University**

Advisor: Professor Dr. Mohammad Ariful Islam

Project: A ceRNA-based multi-omics analysis identifies a prognostic signature in Lung Adenocarcinoma

Overview:

- Conducted a ceRNA-based multi-omics analysis to identify prognostic biomarkers in lung adenocarcinoma, leading to a two-RNA signature comprising ITGB1-DT and IGF2BP1.
- Integrated transcriptomic, survival, mutation, methylation, immune-infiltration, drug-sensitivity, and molecular-docking analyses using TCGA and CPTAC datasets.
- Manuscript under review in **Plos One: See details**

Visiting Graduate Researcher

July 2023 - June 2024

Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka

Advisor: Dr. Shahina Akter

Project: Development of a broad-spectrum epitope-based vaccine against *Streptococcus pneumoniae*

Overview:

- Protein sequence retrieval & epitope prediction, T-cell & B-cell epitope screening, Vaccine construct design with linkers & adjuvants, Physicochemical & immunological analysis, Molecular docking & immune simulation, Codon optimization & in-silico cloning.
- Manuscript Published in **Plos One: See Details**

Undergraduate Research Assistant

October 2022 - June 2023

Advanced Molecular Biology Laboratory, **Jagannath University**, Dhaka

Advisor: Associate Professor Dr. Ahmed Abu Rus

Project: Molecular identification of *Escherichia Coli* from environmental sample

Overview:

- Sample collection & microbial isolation, Culture on selective/differential media (MacConkey, EMB), Biochemical identification of isolates (KIA, SIM, MR-VP, Citrate, Oxidase, Catalase)
- DNA extraction, purification & quantification
- PCR amplification of 16S rRNA gene, Agarose gel electrophoresis & band visualization, Nucleic acid-based microbial identification.

Skills

Wet Lab skills: Microbial culture, enrichment, isolation and plating techniques | Media preparation & aseptic techniques | Biochemical identification: KIA, SIM, MR–VP, citrate, oxidase & catalase tests | DNA extraction, purification & quantification | PCR and 16S rRNA amplification | Agarose gel electrophoresis | Antibigram (Kirby–Bauer)

Computational/Bioinformatics skills: Python | R | Whole Genome Sequence analysis | Rna-seq analysis | Differential expression analysis (DESeq2, edgeR) | Cancer multi-omics analysis (ceRNA networks, mutation, methylation) | TCGA/GEO database analysis | Functional enrichment analysis (GO, KEGG) | Survival and prognostic modelling | Immune-infiltration and tumour-microenvironment analysis | Drug-sensitivity prediction | NCBI sequence-data submission and curation | Single-cell RNA-seq analysis (Seurat) | Primer & Probe design | Automation, data analysis and data visualization

Publications and conferences

Published (1)

Nahian M., Shahab M, **Khan M.R.**, Akash S, Banu TA, Sarkar MH, et al. (2025) Development of a broad-spectrum epitope-based vaccine against *Streptococcus pneumoniae*. PLoS ONE 20(1):e0317216. <https://doi.org/10.1371/journal.pone.0317216>.

Genomic database submissions

Nishan, R. K. (Data submitter). Whole-genome sequencing of ESBL-producing *Escherichia coli* strain S11 isolated from tap water in South Dhaka, Bangladesh. NCBI BioProject [PRJNA1478760](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1478760); GenBank WGS accession [JC](https://www.ncbi.nlm.nih.gov/genbank/JC)[CA](https://www.ncbi.nlm.nih.gov/genbank/CA)[KB](https://www.ncbi.nlm.nih.gov/genbank/KB)[G000000000.1](https://www.ncbi.nlm.nih.gov/genbank/G000000000.1); genome assembly [GCA_059509815.1](https://www.ncbi.nlm.nih.gov/genbank/GCA_059509815.1); BioSample [SAMN61018556](https://www.ncbi.nlm.nih.gov/biosample/SAMN61018556); SRA run [SRR39239575](https://www.ncbi.nlm.nih.gov/sra/SRR39239575), 2026.

In Review (4)

Nishan, R. K., Nahian, M., Islam, M. S., Azad, A. K., & Islam, M. A. A ceRNA-based multi-omics analysis identifies a prognostic signature and predicted drug sensitivities in lung adenocarcinoma. Manuscript under review at **PLOS ONE**.

Ahmed, S., **Nishan, R. K.**, Ferdous, J., Rahman, A., Ferdous, J., Safa, A., Al Mamun, A., Islam, M. S., & Azad, A. K. Whole-genome sequence data of an ESBL-producing *Escherichia coli* from tap water in South Dhaka, Bangladesh. Manuscript under review at **Data in Brief**.

Nishan, R. K., Azad, A. K., & Islam, M. S. Complete blood count profiles by dengue NS1 antigen status among hospital-tested encounters: A retrospective analytical cross-sectional study in Dhaka. Manuscript under review at the **International Journal of Infectious Diseases**.

Acharjee, M., Tayef, M. S., Nahian, M., **Nishan, R. K.**, Sikder, R., & Rahman, N. Burden and trends of antimicrobial resistance among major clinical bacterial pathogens in Bangladesh (2019–2024): A systematic review and meta-analysis. Manuscript under review at the **International Journal of Microbiology**.

In process (4)

Development of a Pan Proteome epitope based Vaccine candidate using bioinformatics tools against Nipah Virus (**First Author**)

Development of a Pan Proteome Vaccine against four Human Pathogenic Ebola viruses using immunoinformatics tools (**First Author**)

Development of a Strip-Based DNA Hybridization Assay for Detection of Selected Gram-Positive Nosocomial Pathogens and Their Resistance and Virulence Markers in Bangladesh (**First author**)

Reframing gut phage therapy as niche-aware ecological engineering: the central role of the mucofilm in immune and metabolic reprogramming (**review article**)

Conferences (4)

Khan, Md. Rasel, Nahian, Md., & Islam, Md. Ariful. (2024). mRNA–miRNA–lncRNA interaction network in different stages of lung adenocarcinoma. Three-minute oral presentation, *38th BSM Annual International Conference 2024*. Abstract 3MP-C6, pp. 94–95. **Presenting author**.

Sharmile, N., Zaman, F. B., **Khan, Md. Rasel**, & Islam, Md. Ariful. (2024). Isolation and molecular characterization of bacterial strains from wastewater samples. *37th BSM Annual Conference 2024*. Abstract ENR-P23, p. 70. **Presenting author. (1st runner up poster presentation prize)**

Sathy, S. A., **Khan, Md. Rasel**, Nahian, Md., Mia, R., Afroz, F., & Azad, A. K. (2024). Isolation, identification and antibiotic sensitivity testing of water-borne bacteria from different tap water of Dhaka city, Bangladesh. *37th BSM Annual Conference 2024*. Abstract FAR-P27, p. 85.

Mazumder, L., Nahian, Md., Metrika, M. A., Gulshan, R., Fahiha-Rahman, **Khan, Md. Rasel**, Islam, M. A., & Begum, S. (2023). Designing of novel epitope-based vaccine candidates against dengue virus isolates from Bangladesh. *36th BSM Annual Conference*. Abstract GEBP-15, p. 82.

Test Scores

IELTS: Overall - **7.5** (Listening 8, Reading 8.5, Writing 6.5, Speaking 6.5)

November 2025

Awards & Honors

1st Runner-up poster presentation 37th BSM Annual Conference

January 2024

National Science & Technology (NST) Fellowship 2023-24

October 2023

SSC Merit Scholarship Dinajpur Education Board

June 2015

Best Reader Award, SEQAEP (Secondary Education Quality & Access Enhancement Project), Class 7

November 2011

Training Experience

RNA-Seq with Bioconductor in R (Datacamp)

May 2025

Introduction to R (Datacamp)

March 2024

Whole genome sequencing of bacterial genomes tools and applications (Coursera)

May 2023

Hands-on Training & workshop on RT-PCR based Molecular Diagnostic Technique, Graduate Microbiologist Society (GMS) Bangladesh

June 2022

Volunteering Experience

Volunteer, Seminar on Public Health & AI Department of Microbiology, Jagannath University

May 2025

Volunteer, Tour Management – Microbiology Department Excursion

November 2022

References

Academic references and contact details are available upon request.